VARSHINI GUTTULA

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OBJECTIVE

A self-motivated and quick learner with a strong analytical mindset and a passion for technology. Adept at problem-solving and delivering innovative solutions through technical expertise.

SKILLS

Technical Skills:

- Languages - Python
- Web Technologies - HTML, CSS, React JS
- Tools & Frameworks: TensorFlow, Scikit
- UI/UX: Figma, Canva
- Database – SQL (basics)

Non-Technical Skills:
Communication, Leadership, Team player, Adaptability, Problem-solving

WORK EXPERINCE

AI Intern **July 2021 - December 2025**
Kernn Automations - India

- Built a resume classification model using TF-IDF vectorization and Logistic Regression on ~1,000 labelled resumes
- Implemented end-to-end NLP pipeline: text preprocessing, feature extraction, train/test splitting, and model evaluation using precision, recall, and F1-score
- Serialised trained model and vectorizer using joblib for reuse without retraining
- Identified next-phase improvements including transformer-based embeddings and FastAPI deployment

EDUCATION

Master in Computer Science **March 2026 – Current**
University Of Wollongong

Bachelor of Technology in Computer Science and Engineering - AI/ML **September 2021 - June 2025**
Malla Reddy University
CGPA: 8.8

PROJECTS

CRNN- A Novel Hybrid Approach for Sign Language Gesture Recognition | OpenCV, NumPy, TensorFlow (CNN, LSTM), Data Visualization (Matplotlib), Scikit-learn, Image Data Generator

- Engineered a CRNN model leveraging CNNs for spatial feature extraction and RNNs for temporal dynamics, **improving recognition accuracy by 15% and robustness by 25%**
- **Achieved a 40% reduction in processing latency** for real-time sign language gestures, enhancing communication efficiency
- **Acceptance in 16th ICCCNT 2025 – IEEE Conference, IIT Indore, Paper ID: 5109**

Gesture-Controlled Switches for Smart Home Automation Using Acoustic Sensors | Arduino Uno R3, Acoustic Sensors, Embedded C, Signal Processing, IoT

- Designed a smart home switch enabling contactless operation through gestures, enhancing convenience and efficiency
- Implemented acoustic signal processing techniques to accurately detect and interpret user gestures
- Optimized embedded firmware to **reduce latency by 20%**, ensuring seamless real-time interaction

LEARNABILITY: Detection of Learning Disorders Using Machine Learning | Python, Machine Learning (Random Forest), Pandas, Scikit-learn
Professor, A. S. K. A et al. (2023). Indian Scientific Journal of Research in Engineering and Management, 07(12), 1–11. https://doi.org/10.55041/ijsrem27568

- Developed a machine learning system using Random Forest for accurate detection of learning disorders, primarily dyslexia
- **Achieved 85% accuracy** in detecting learning disorders, primarily dyslexia.

CERTIFICATIONS

- **NPTEL - Deep Learning:** Proficiency in advanced deep learning concepts and architectures.
- **IBM - Introduction to Machine Learning Specialization:** Foundational knowledge in ML and real-world applications.
- **Google Cloud Computing Foundations & Generative AI**
- **Java Programming Fundamentals- Coursera**
- **AWS Academy Graduate - AWS Academy Machine Learning Foundations:** Comprehensive understanding of ML concepts and cloud integration.

ACHIEVEMENTS

- Runner-up in Nav-Nirman Hackathon (2023), I&E Cell MRUH, for "Easy Shop", an innovative cafeteria management solution
- Vice President of E-Cell – MRUH, Ranked 36 among 300+ teams in National Entrepreneurship Challenge (NEC), IIT Bombay
- Head of Social Media Committee - MRUH

INTERESTS

Coding, Graphic Designing, Public Speaking, Performing arts